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Research Paper

A Spectral–Geometric Tensor Framework Beyond Hartree–Fock and Kohn–Sham Closure

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Abstract

A Spectral–Geometric Tensor (SGT) framework is developed as an operator-based reduced description of selected atomic ionisation observables. The construction starts from channel-resolved regular families of self-adjoint operators, isolated finite-rank spectral sectors, and their Riesz projectors. The parameter manifold is mapped by the projector into a Grassmannian of reduced quantum-state subspaces. On this image, the quantum geometric tensor defines the metric and Berry curvature, and the metric generates the Levi–Civita, Riemann, Ricci, and scalar-curvature hierarchy. This geometry is distinguished explicitly from the leading energetic closure: for ordinary channels, the numerical benchmark retains an anchor-normalised reduced-gap law, whereas the near-degenerate $4s/d$ sector is represented by a reduced 2×2 mixing Hamiltonian. The gap is a spectral scalar and becomes geometrically informative only together with derivative couplings; curvature invariants enter the first omitted correction layer rather than the reported leading-order values. Benchmarking on H , He , Li , Be , B , C , N , O , F , Ne , Cr , Cu , and Pd yields $RMSE = 0.76097$ eV, $RMSE^* = 5.19146\%$, $MAE = 0.43628$ eV, and $MAPE = 2.97794\%$, with residual support confined to the $2p$ sector. The result is a spectrally intrinsic leading closure embedded in an explicit geometric hierarchy, with the boundary between implemented quantities and prospective curvature corrections stated precisely.

Keywords

Spectral theory; Self-adjoint operators; Quantum geometric tensor; Riesz projector reduction; Electronic structure theory

Introduction

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Electronic-structure theory has long been organised around wavefunctions, orbitals, and densities as primary objects [4], [5], [6]. That architecture has been remarkably successful, but it also fixes the mode of closure at the foundation of the theory: Hartree–Fock through orbital mean-field structure, and Kohn–Sham through density-based effective closure [7], [8], [9], [10], [11], [12]. In both cases, the observable architecture is subordinated to density- or orbital-mediated reconstruction. A different route is developed here. The primitive datum is neither orbital nor density, but a regular family of self-adjoint operators together with an isolated spectral sector stable under admissible deformation [1], [2], [3]. Observable structure is then generated through projector reduction, bundle organisation, and tensorial response [13], [16], [24].

This programme must be placed within the established nonadiabatic and quantum-geometric literature. Tully’s surface-hopping construction made nonadiabatic coupling vectors operational in molecular dynamics, and subsequent reviews have systematised their use in mixed quantum–classical propagation [17], [18]. Independently, the quantum geometric tensor and its Riemannian content were established by Provost and Vallée [16]. Finite-rank Abelian and non-Abelian versions were later formulated for degenerate sectors [19]. Recent work has related transition matrix elements to Riemannian response, developed covariance-based descriptions of quantum-state manifolds, and extended the construction toward Cartan-type connections, torsion, and curvature [20], [21], [22]. Accordingly, neither the quantum geometric tensor nor the emergence of Riemannian or Cartan geometry is claimed here as new.

The specific contribution of the present work is the observable layer placed after this established geometry. A Riesz projector first selects an isolated atomic channel; the compressed spectral quantities $\{\Delta_g, \mathbb{H}_{sd}\}$ then define a channel-resolved energetic descriptor, and an explicit map Φ_g sends that descriptor to the first ionisation observable I_1 . Thus the present proposal is not another derivation of quantum-state geometry. It is a test of whether projector-defined spectral sectors can support an anchor-normalised, regression-free observable closure, while the metric and gauge curvatures identify the invariant structure omitted by that leading closure.

The terminology “reduced spectral manifold” is used in a precise sense. For each channel g , M_g is the parameter manifold of a family of quantum operators and states. The projector defines a smooth map

$$P_g : M_g \longrightarrow \text{Gr}_{r_g}(\mathcal{H}_g), \quad \lambda \mapsto \text{Ran } P_g(\lambda),$$

and the reduced spectral manifold is the regular image of this map, locally identified with M_g when the differential of P_g has constant rank. The projector does not create the external control variables. It converts them into a family of physical quantum subspaces. Conversely, projected expectation values

$$x_g^\alpha(\lambda) := \text{Tr}[P_g(\lambda) O_\alpha]$$

provide local coordinates whenever the Jacobian $\partial_\mu x_g^\alpha$ has full rank. Atomic elements are discrete samples $\lambda_g(Z) \in M_g$, not differentiable coordinates by themselves.

Three objectives are therefore separated. First, the paper states the operator-theoretic hypotheses under which isolated finite-rank sectors define smooth bundles. Secondly, it writes the complete geometric hierarchy $Q \rightarrow (g, \Omega) \rightarrow (\Gamma, R, \text{Ric}, \mathcal{R})$, including its finite-rank interpretation and its relation to nonadiabatic coupling vectors. Thirdly, it implements a leading atomic closure through the compressed gap Δ_g or the mixing matrix $\mathbb{H}_{sd}$, and states explicitly which geometric invariants are retained only at the next correction order. This separation is essential: the gap is spectrally intrinsic, but it is not a geometric tensor by itself.

First ionisation energies provide the test domain. In ordinary channels the leading observable depends on a compressed-gap ratio defined after projector reduction. In the near-degenerate $4s/d$ sector it is generated by a reduced 2×2 spectral mixing problem. The benchmark therefore tests a reduced closure principle, not a claim that scalar curvature has already been measured for each atom. Geometry organises the state manifold and constrains systematic corrections; the reported leading numerical values use only the explicitly identified spectral descriptors.

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Section snippets

Regular operator families, reduced sectors, and atomic instantiation

Definition 2.1 Admissible deformation manifold

Let M be a connected smooth finite-dimensional manifold. A point

$$\lambda = (\lambda^1, \dots, \lambda^k) \in M$$

is called an admissible deformation parameter. The coordinates λ^μ encode reduced spectral control variables through which a fixed isolated sector evolves. ...

Definition 2.2 Spectral structural representation

Let $\mathcal{H}$ be a separable complex Hilbert space, and let

$$H : M \rightarrow \mathcal{L}_{\text{sa}}(\mathcal{H}), \quad \lambda \mapsto H(\lambda),$$

be a family of self-adjoint operators on a common dense domain $\mathcal{D} \subset \mathcal{H}$. The pair $(\mathcal{H}, H)$ is called a spectral structural representation. ...

Remark 2.3 Regularity regime

Regularity means precisely what is required to guarantee local ...

...

Benchmark protocol

Once the reduced ionisation law has been established within the spectral–tensorial framework, the remaining task is quantitative: to determine how much of the observed first-ionisation structure is already resolved by the leading reduced closure, to identify the sectors in which scalar compression is sufficient, and to isolate the first regimes in which genuinely multicomponent or higher-order tensorial effects become active.

The benchmark set is chosen as

$$\mathcal{B} = \{H, He, Li, Be, B, C, N, O, F, Ne, Cr, Cu, Pd\}.$$

...

Conclusion

A reduced spectral–geometric framework has been established for selected atomic first-ionisation observables. The construction starts from regular self-adjoint operator families and isolated spectral sectors, proceeds through Riesz-projector reduction and bundle geometry, and culminates in a local tensorial structure induced by the quantum geometric tensor [1], [13], [24]. The decisive conceptual point is that the observable is not built from orbitals or densities taken as primitive objects. ...

Uncited references

[14], [23], [28], [34], [35] ...

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CRedit authorship contribution statement

Cesar A. de Mello: Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. ...

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Cesar A. de Mello reports writing assistance was provided by Cosmo Physics Labs. If there are other authors, they declare that they have no known competing financial interests or personal

relationships that could have appeared to influence the work reported in this paper.

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References (36)

T. Kato

Perturbation Theory for Linear Operators

(1995)

M. Reed *et al.*

Methods of Modern Mathematical Physics, Vol. I: Functional Analysis

(1980)

M. Reed *et al.*

Methods of Modern Mathematical Physics, Vol. IV: Analysis of Operators

(1978)

T. Helgaker *et al.*

Molecular Electronic-Structure Theory

(2000)

A. Szabo *et al.*

Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory

(1996)

R.G. Parr *et al.*

Density-Functional Theory of Atoms and Molecules

(1989)

P. Hohenberg *et al.*

Inhomogeneous Electron Gas

Phys. Rev. (1964)

W. Kohn *et al.*

Self-Consistent Equations Including Exchange and Correlation Effects

Phys. Rev. (1965)

J.P. Perdew *et al.*

Jacob's Ladder of Density Functional Approximations for the Exchange-Correlation Energy

AIP Conf. Proc. (2001)

J.P. Perdew *et al.*

Generalized Gradient Approximation Made Simple

Phys. Rev. Lett. (1996)



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